

Answer Key — The World of Microorganisms

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 (c) **viruses** — viruses are acellular and need a living host to reproduce.

Q2 (b) **protozoa** — both are single-celled protozoa.

Q3 (a) **unicellular fungus** — yeast is a single-celled fungus.

Q4 (c) **algae** — algae are green and photosynthesise.

Q5 (b) **False** — they need a microscope to be seen.

Q6 **living being** — organism means living being.

Q7 **spiral** — bacteria come in spherical, rod, comma and spiral shapes.

Short answer

Q8 Bacteria (Lactobacillus), Protozoa (Amoeba), Fungi (yeast), Algae (Spirulina).

Q9 Viruses are acellular (not made of cells) and multiply only inside a living host cell.

Q10 Water, soil, air, hot springs, snow, inside the bodies of plants and animals (any two).

Long / application

Q11 Pond water: protozoa (Amoeba, Paramecium) and algae. Soil suspension: fungi (mould), bacteria, algae. A microscope is essential because microbes are smaller than the eye can see; it magnifies them 100–400 times.

Q12 (i) Amoeba — unicellular, protozoa; (ii) bread mould — multicellular, fungi; (iii) Lactobacillus — unicellular, bacteria; (iv) yeast — unicellular, fungi.
